



2.19 Chloroplast

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What you will learn

- Structure and function of chloroplast
- Parts of the chloroplast used in **photosynthesis**



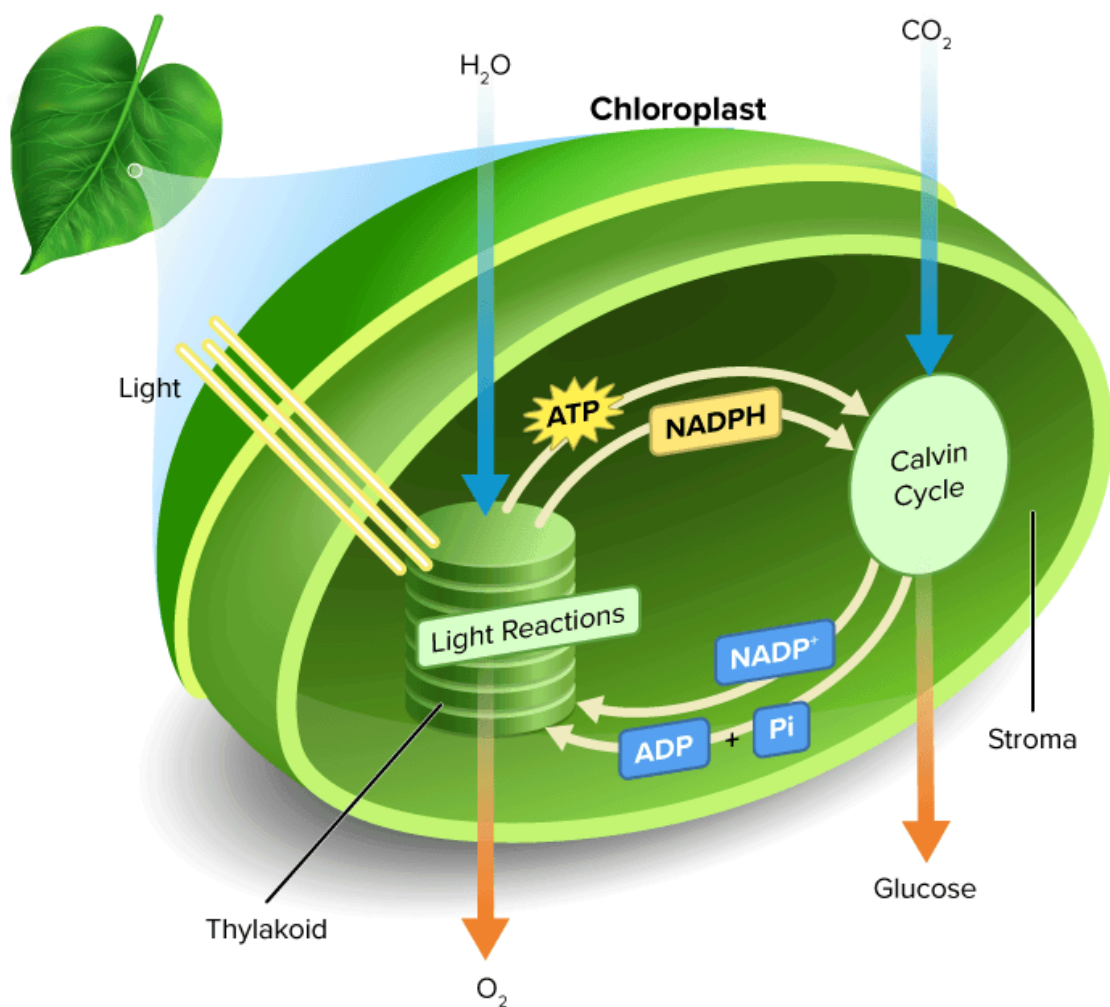
[Figure 1]

The chloroplast is the site of photosynthesis. Part of the photosynthesis reactions occurs in an internal membrane within the **organelle**. The chloroplast contains many of these internal membranes, making photosynthesis very efficient. These internal membranes stack on top of each other, just like a stack of pancakes.

Stages of Photosynthesis

Photosynthesis occurs in two stages, which are shown in the **Figure** below.

1. Stage I is called the **light reactions**. This stage uses **water** and changes light **energy** from the sun into chemical energy stored in **ATP** and **NADPH** (another energy-carrying molecule). This stage also releases oxygen as a waste product.
2. Stage II is called the **Calvin cycle**. This stage combines carbon from carbon dioxide in the air and uses the chemical energy in ATP and NADPH to make **glucose**.



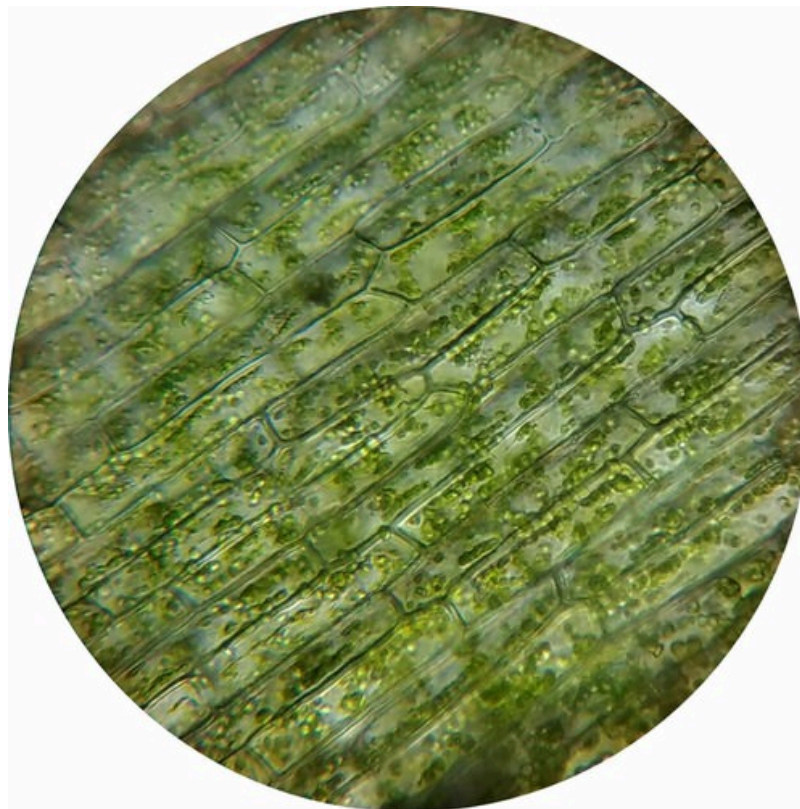
[Figure 2]

The two stages of photosynthesis are the light reactions and the Calvin cycle. Do you see how the two stages are related?

Before you read about these two stages of photosynthesis in greater detail, you need to know more about the chloroplast, where the two stages take place.

Photosynthesis, the process of turning the energy of sunlight into **glucose**, is divided into two basic sets of reactions, known as the light reactions and the Calvin cycle, which uses carbon dioxide. As you study the details in other concepts, refer frequently to the **chemical equation** of photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Light Energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Photosynthesis occurs in the chloroplast, an organelle specific to plant **cells**.

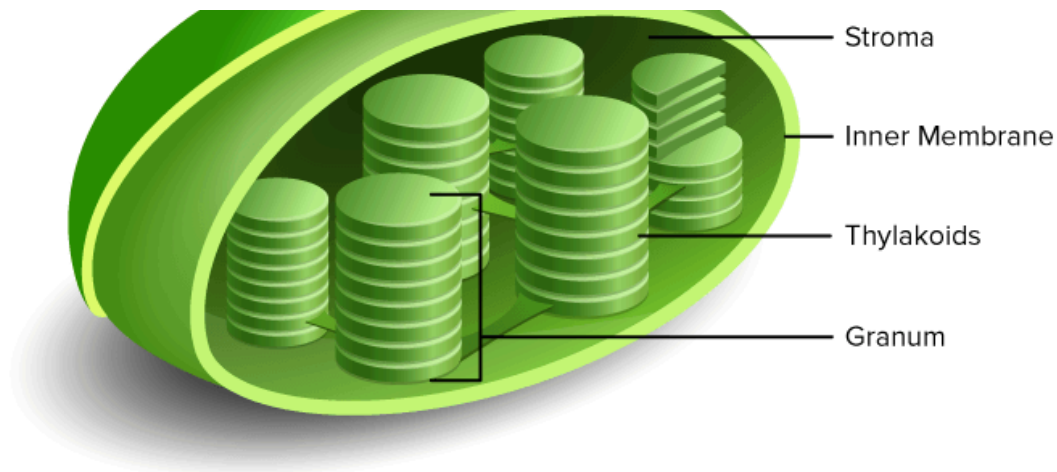
If you examine plant cells, shown in the **Figure below**, under a **microscope**, you will see within each cell dozens of small green ovals. These are **chloroplasts**, the organelles which conduct photosynthesis in plants and **algae**. Chloroplasts closely resemble some types of **bacteria** and even contain their own circular **DNA** and **ribosomes**. In fact, the **endosymbiotic theory** holds that chloroplasts were once independently living bacteria (prokaryotes). So when we say that photosynthesis occurs within chloroplasts, we speak not only of the organelles within plants and algae but also of some bacteria – in other words, virtually all photosynthetic **autotrophs**.



[Figure 3]

Plant cells under a microscope. Many green chloroplasts are visible.

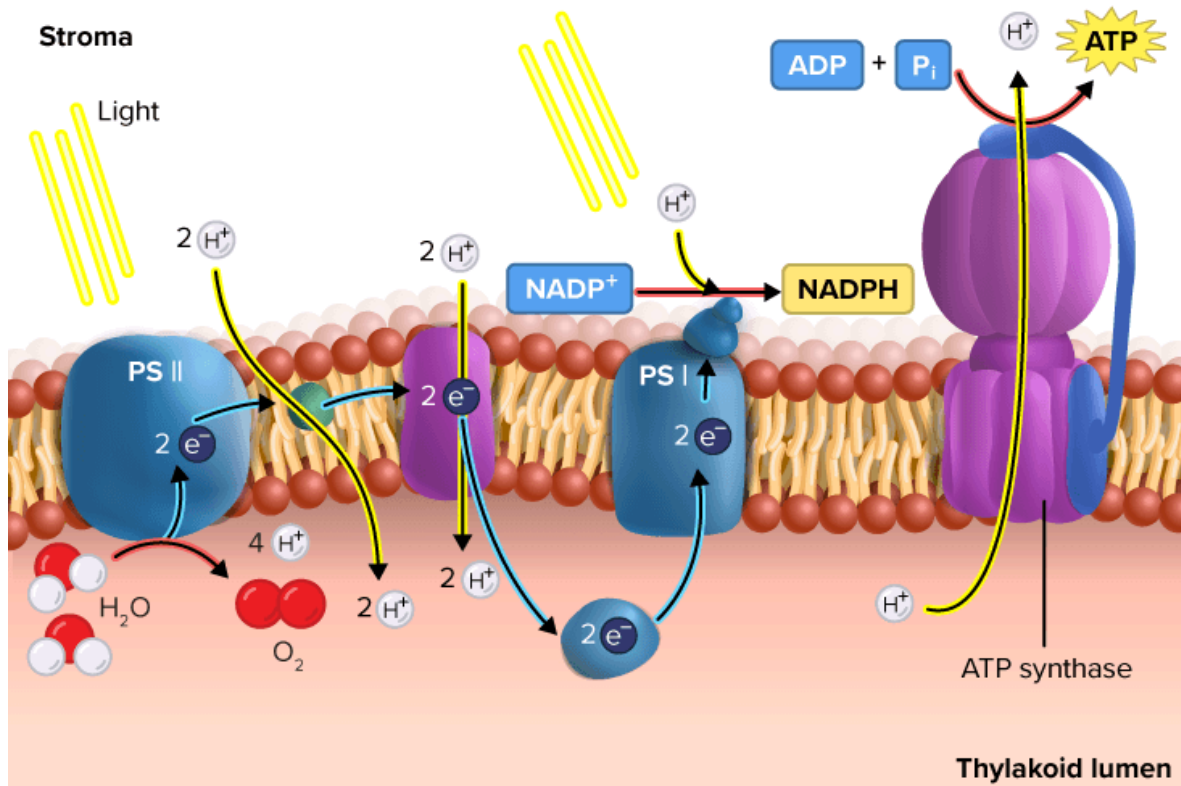
Each chloroplast contains neat stacks called **grana** (singular, granum). The grana consist of sac-like membranes, known as **thylakoid** membranes. These membranes contain **photosystems**, which are groups of molecules that include **chlorophyll**, a green pigment. The light reactions of photosynthesis occur in the thylakoid membranes. The **stroma** is the space outside the thylakoid membranes. This is where the reactions of the Calvin cycle take place. In addition to **enzymes**, two basic types of molecules - pigments and **electron carriers** – are key players in this process and are also found in the thylakoid membranes.



[Figure 4]

A chloroplast consists of thylakoid membranes surrounded by stroma. The thylakoid membranes contain molecules of the green pigment chlorophyll.

Electron carrier molecules are usually arranged in **electron transport chains** (ETCs). These accept and pass along energy-carrying electrons in small steps (**Figure below**). In this way, they produce ATP and NADPH, which temporarily store chemical energy. Electrons in transport chains behave much like a ball bouncing down a set of stairs – a little energy is lost with each bounce. However, the energy “lost” at each step in an **electron transport chain** accomplishes a little bit of work, which eventually results in the synthesis of ATP.



[Figure 5]

This figure shows the light reactions of photosynthesis. This stage of photosynthesis begins with photosystem II (PS II) (so named because it was discovered after photosystem I (PS I)). Find the two electrons ($2e^-$) in photosystem II, and then follow them through the electron transport chain (also called the electron transfer chain) to the formation of NADPH.

Where do the hydrogen ions (H^+) come from that help make ATP?



Summary

- Photosynthesis occurs in two stages, the light reactions and the Calvin cycle.
- Photosynthesis occurs in the chloroplast, an organelle specific to plant cells.
- The light reactions of photosynthesis occur in the thylakoid membranes of the chloroplast.
- Electron carrier molecules are arranged in electron transport chains that produce ATP and NADPH, which temporarily store chemical energy.

Review

1. Describe the chloroplast's role in photosynthesis.
2. Explain how the structure of a chloroplast (its membranes and thylakoids) makes its function (the chemical reactions of photosynthesis) more efficient.
3. Describe electron carriers and the electron transport chain.



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